

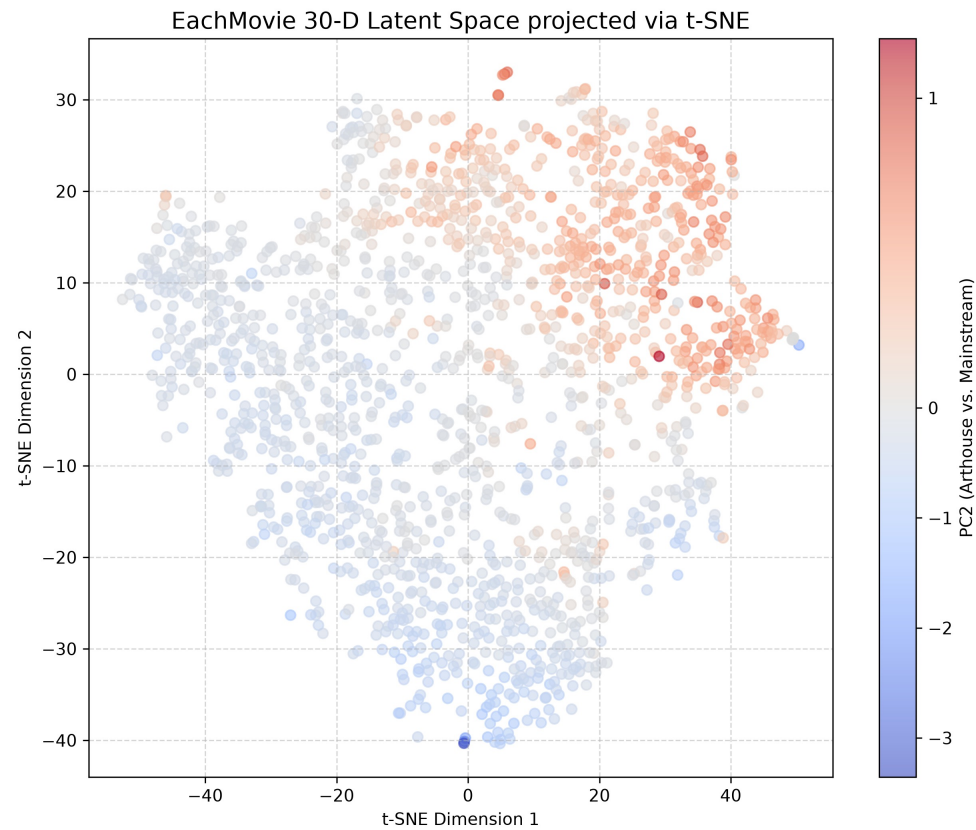
EachMovie: archaeology of the/a first latent recommender system

ACM RecSys 1996→2026
Past, Present, and Future

John DeTreville / DEC Systems Research Center (ret'd) / 2026-09-30 / Minneapolis, MN, USA

eachmovie.github.io

No t-SNE charts (not that kind of talk)



The 1990s

- Computers were about 1000× slower than now.
- The Web had only just begun.
- There were *almost no recommender systems!*
- And there were *no latent recommender systems* —*yet!*
- “We could change the world . . .”
 - . . . **NOT !!!** (At first, anyway. Read the paper.)

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Guidance from Roger Ebert

(Very famous movie critic, *Chicago Sun-Times*)

- *“The star rating system is relative, not absolute.”*
- *“When you ask a friend if **Hellboy** is any good, you’re not asking if it’s any good compared to **Mystic River**, you’re asking if it’s any good compared to **The Punisher** . . .*
- *“If **Superman** is four, then **Hellboy** is three and **The Punisher** is two.”*

Roger Ebert, “Shaolin Soccer”, 2004

EachMovie (1995–97)

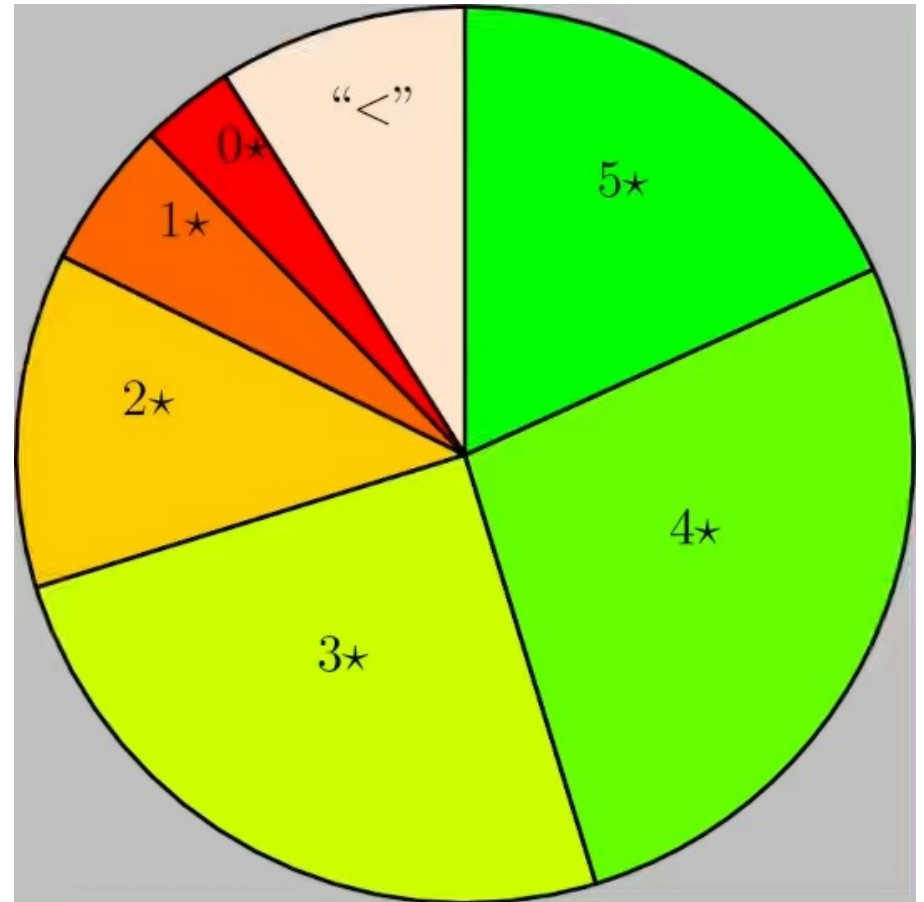
*“If you can name some movies you've liked . . .
we can name some others you might like too!”*

→US6078740A

- Grew to 1,628 movies, 72,916 users, 2,811,983 votes.
- The vote matrix was 97.63% sparse.
- Votes ranged from 0★ to 5★, plus Sounds Awful.

Vote distribution and missing votes

- EachMovie's recorded votes were 9% **Sounds Awful** (" $<$ "), then 3% **0★**, 5% **1★**, 12% **2★**, 25% **3★**, 27% **4★**, and 18% **5★**.
- But 97.63% are missing from the vote matrix.
 - What to do, what to do?



Unfilled objective function

$$\sum_{(i,j)} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

K was 30

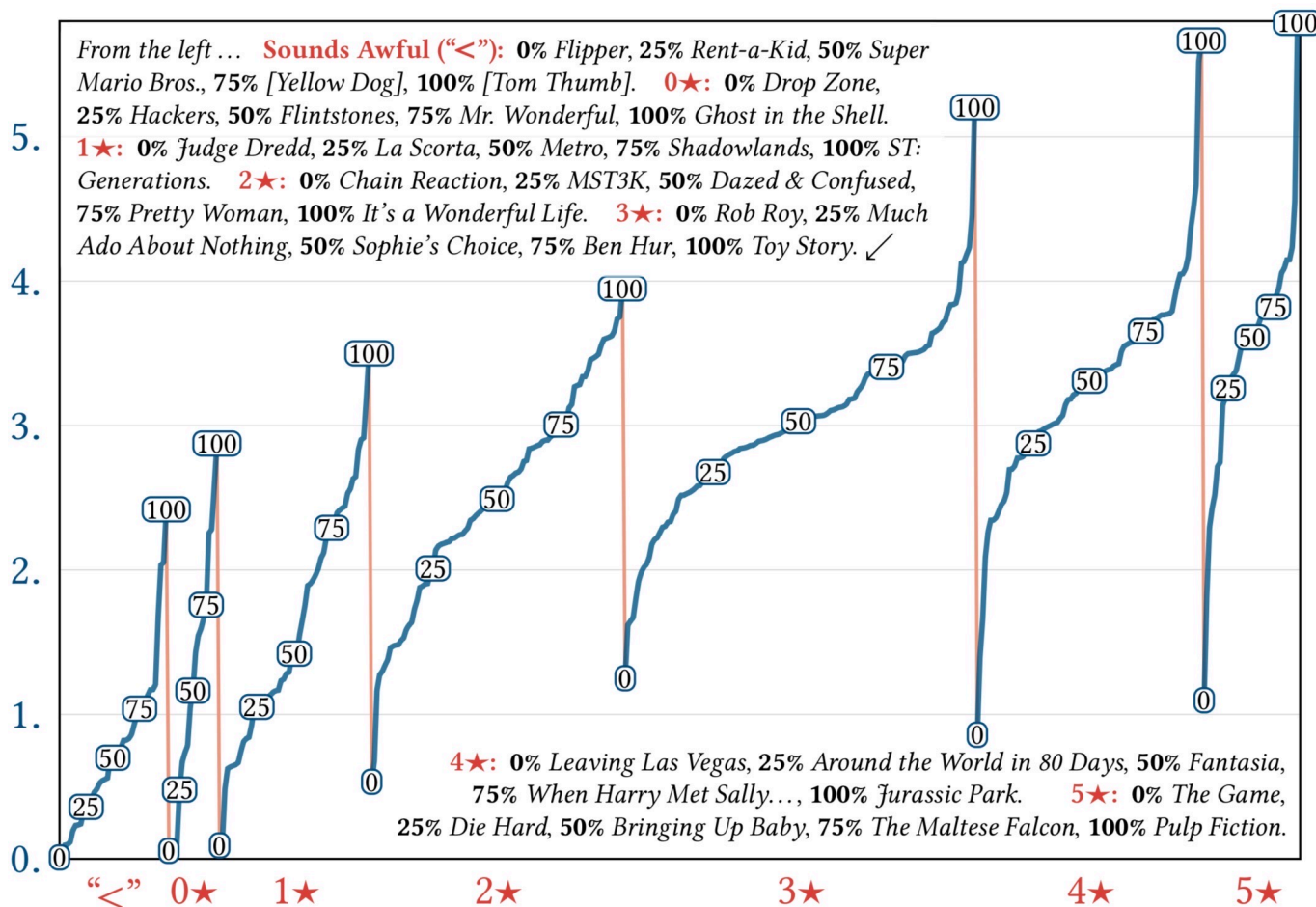
$$\sum_{(i,j) \in \Omega} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

$$\sum_{(i,j) \in \Omega} W_{ij} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

$$\sum_{(i,j) \in \Omega} W_{ij} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2 + \lambda \left(\sum_i \|\vec{u}_i\|^2 + \sum_j \|\vec{v}_j\|^2 \right)$$

λ was 2.0 (or 0.08)

Giraffe chart



User reactions

From: Anonymous

“Congratulations on your site, now I am using it to decide which movies to go to on weekends :)”

From: DB

“Amazing, I am going to have to get to the video store to check out some predictions.”

From: CC

“I love the service. I think that this is one of the coolest and most useful things I’ve ever found on the Internet. I give you my seal of approval.”

From: AK

“I enjoyed using your service and was pleasantly surprised to see some off-beat films listed -- *‘I Shot Andy Warhol’* and *‘Eat Drink Man Woman’* for example.”

From: RP

“This service is a brilliant idea and one that will indeed begin to define both the Web and marketing in the years to come.”

From: ??

“Just wanted to slip out a quick note- this is a greatly enjoyable site! It has lead us (my wife and myself) to some really great movie choices that we otherwise would have overlooked, but I guess that was the idea, wasn't it :)”

From: MDM

“I’ve been using EachMovie for about a month now and, I have to admit, you are pretty good in your predictions. Not all the time, of course you’re not God, but much of the time you're close.”



Image: CC BY-SA 2.0 / Sound Opinions

Roger Ebert (1942–2013)

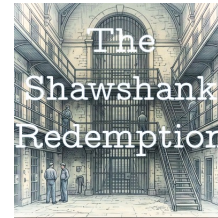


*“[EachMovie] proposes to suggest movies you might enjoy based on your opinions of movies you have seen. [...] To my absolute astonishment, the program evaluated my ratings and advised me I would probably enjoy Pather Panchali, Before the Rain, Land and Freedom, Chungking Express, and Red. This was **uncannily accurate** [...] The program was so much on the money, indeed, that I pictured a Wizard of Oz-type figure hiding behind the curtains, manually inserting suggestions, so I logged on again using a different address and password, and this time the program’s recommendations included the splendid choices Theremin, Fargo, and Crumb. I am forced to concede that EachMovie works ... for me, anyway.”*

Roger Ebert, Yahoo! Internet Life 2, 4 (Sep. 1996)

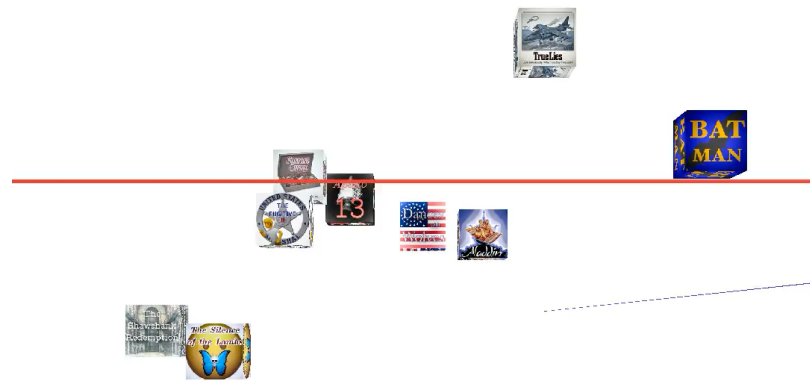
EachMovie's "Top Ten"

10. *Aladdin* (228K★)
9. *The Shawshank Redemption* (231K★)
8. *Forrest Gump* (235K★)
7. *The Silence of the Lambs* (242K★)
6. *True Lies* (263K★)
5. *Batman* (263K★)
4. *The Fugitive* (266K★)
3. *Pulp Fiction* (304K★)
2. *Dances With Wolves* (320K★)
1. *Apollo 13* (329K★)



All third-party image sources for these crudely drawn derived works are in the public domain or otherwise freely available in the United States. For further details, see [Coyote vs. ACM \(2026\)](#).

PC 1&2&3→2&3&4 projected through 30-space



(pc1, pc2) → (pc2, pc3)



EachMovie (old)

- EachMovie shut down in late 1997.
- The EachMovie dataset was available to researchers until 2004 and was widely used.
- Top social-media post about EachMovie: *“Where can I find the original EachMovie dataset? (for a research paper I’m working on)”*



If you can name some movies you've liked...
we can name some others you might like too!

Roger Ebert calls us "**uncannily accurate**"! Want to try us now? Please enter your:

E-mail address:

Password: (optional)

[LOG IN](#)

(To provide personal predictions, we keep your data separate from everyone else's. We use your e-mail address to identify you, and an optional password to keep your data private.)

Return to EachMovie as often as you like; just bookmark any of our pages. Come back each week to check out the new releases and tell us about the movies you've seen!

Our current listings highlight recent film and video releases in the United States.

Summer comes to Palo Alto! It must be here, since we've just experienced 1997's first **power outage**! Our apologies for Sunday's interruption of service!

EachMovie is shutting down! It's sad but true. We started EachMovie one year ago as a **research project** for Digital Equipment Corporation. Now the project's **over**! EachMovie will stay up through **the end of February**, although we're actively looking for a **buyer**. In the meanwhile, thanks for using **EachMovie**!

Microsoft Internet Explorer users! Using EachMovie with MSIE 3.0 and 3.01 can cause Windows 95 to run out of resources. For now, switch to Netscape Navigator if you have a problem.

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EachMovie (new)

- A resurrected EachMovie website is planned for researchers and historians.
- Check the EachMovie blog at eachmovie.github.io for updates.
- *Careful: the website may be running on an old Mac Mini in my garage.*



Here are some predictions for old films and videos. Rating more movies improves your predictions; you've rated **465** so far.

★★★★★	? unseen	◇	Interview with the Vampire	IMDb
★★★★★	? unseen	◇	Beauty and the Beast	IMDb
★★★★★	? unseen	◇	The English Patient	PR IMDb
★★★★★	? unseen	◇	To Kill a Mockingbird (1962)	IMDb
★★★★★	? unseen	◇	All About Eve (1950)	IMDb
★★★★★	? unseen	◇	Some Like it Hot (1959)	IMDb
★★★★★	? unseen	◇	12 Angry Men (1957)	IMDb
★★★★★	? unseen	◇	The Adventures of Robin Hood (1938)	IMDb
★★★★★	? unseen	◇	The Manchurian Candidate (1962)	IMDb
★★★★★	? unseen	◇	A Grand Day Out	IMDb

[PRESS HERE](#) (see options below)

- List more of these old films and videos (1153 remaining)
- ☐ See predictions for :
- ☐ Find titles with the word
- ☐ Review my votes so far and add comments ...

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Epigraph / epilogue

*I have heard the mermaids singing, **each to each**.
I do not think that they will sing to me.*

— “The love song of J. Alfred Prufrock,” T.S. Eliot, 1917

- So, I figured, let's **do** something about that!
And that's the story of EachMovie.

Past, Present, and FUTURE

- **PAST:** EachMovie factored the vote matrix to expose the latent reality underneath. Users **wanted** to use EachMovie. They sought it out. They gladly shared their tastes & souls with it—and it returned the favor by **letting them drive**.
- **PRESENT:** Those carefree halcyon days of the 1990s are long gone. Now our modern dystopia dials the **doomscrollamine** ^[1] all the way up to 11.
- **FUTURE:** “There are two paths we can go by.” ^[2] And there’s still time to change the road we’re on.

Extended Summary

- a) EachMovie let users drive.
- b) Doomsrollamine is bad. [1]
- c) **There's still time to change
the road you're on.** [2]

QUESTIONS?

[1] Andrew Marltan, 2026.

[2] Robert Plant, 1971.

EachMovie's Top Ten Movies

"Across 30 Dimensions"

—A Music Video—

by John DeTreville, 2026